Gaosheng Liu

Work permit: Italian | **Date of birth:** 22/01/1995 | **Place of birth:** Anhui, China |

Nationality: Chinese | **Gender:** Male | **Phone number:** (+86) 18822321576 (Mobile) |

Email address: lgselite@163.com | **Website:** https://lgs001elite.github.io/ | **LinkedIn:** https://www.linkedin.com/in/gaosheng-liu-a99340207/ |

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ABOUT ME

I am a postdoc researcher at the University of Trento. Throughout my academic career, my research has primarily focused on communication-related challenges in battery-free IoT systems. During my Master's, I investigated secure communication methods, and throughout my PhD, I addressed communication challenges in battery-free IoT sensing systems. My broad research interests extend to IoT protocols, application deployment, and interdisciplinary research. I have a strong publication record, including papers accepted by prestigious venues such as TMC 2024, APNet 2024, TrustCom, and a Best Paper Award at IPCCC 2023.
Additionally, I have one paper under submission for SIGCOMM 2025. I am a highly motivated researcher who has independently completed my PhD within four years, primarily collaborating only with my supervisor. I am confident that joining a reputable platform would allow me to enhance my research productivity and impact further. Besides my academic achievements, I bring two years of industry experience as a network software engineer at Huawei. This role significantly strengthened my practical programming skills, debugging abilities, problem-solving capabilities, and familiarity with industry-level software development. My combined academic and industry experiences have equipped me with robust critical thinking, innovation capabilities, academic writing expertise, strong programming skills, and practical communication abilities. I am also qualified and enthusiastic about supervising Master's and PhD students, assisting them with idea generation, problem-solving, debugging, and academic writing. I am confident in contributing effectively as a researcher and mentor. In my personal life, I enjoy hiking, cycling, and playing badminton, particularly appreciating nature's beauty and peaceful surroundings. I am also passionate about exploring diverse cuisines—especially seafood, street food, and Chinese dishes—and enjoy quality time with family, particularly spending leisure moments with my wife.

EDUCATION AND TRAINING

08/01/2021 – 08/01/2025 Amsterdam, Netherlands

PHD RESEARCHER Vrije Universiteit Amsterdam

Website https://vu.nl/nl | **Level in EQF** EQF level 8

01/09/2016 – 01/01/2019 Tianjin, China

MASTER Tianjin University

Website https://www.tju.edu.cn/english/index.htm | **Level in EQF** EQF level 7

01/09/2011 – 01/06/2015 Tianjin, China

BACHELOR Tianjin University of Technology and Education

Website https://www.tute.edu.cn/ | **Level in EQF** EQF level 6

PUBLICATIONS

2025

FreeBeacon: Efficient Communication and Data Aggregation in Battery-Free IoT (Under submission SIGCOMM2025)

To improve sustainability, Internet-of-Things (IoT) is increasingly adopting battery-free devices powered by ambient energy scavenged from the environment. The unpredictable availability of ambient energy leads to device intermittency, bringing critical challenges to device communication and related fundamental operations like data aggregation. We propose FreeBeacon, a novel scheme for efficient communication and data aggregation in battery-free IoT. We argue that the communication challenge between battery-free devices originates from the complete uncertainty of the environment. FreeBeacon is built on the insight that by introducing just a small degree of certainty into the system, the communication problem can be largely simplified. To this end, FreeBeacon first introduces a small number of battery-powered devices as beacons for battery-free devices. Then, FreeBeacon features protocols for battery-free devices to achieve robust interaction with the beacon and to perform communication efficiently following customized schedules that implement different data aggregation schemes. We evaluate FreeBeacon with extensive simulation studies and prototype-based experiments. Results show that FreeBeacon can consistently achieve an order of magnitude data aggregation time reduction when compared with the state-of-the-art approaches.

Gaosheng Liu, Kasim Sinan Yildirim, and Lin Wang

2024

Data on the go: Seamless data routing for intermittently-powered battery-free sensing

The rising demand for sustainable IoT has promoted the adoption of battery-free devices intermittently powered by ambient energy for sensing. However, the intermittency poses significant challenges in sensing data collection. Despite recent efforts to enable one-to-one communication, routing data across multiple intermittently-powered battery-free devices, a crucial requirement for a sensing system, remains a formidable challenge. This paper fills this gap by introducing Swift, which enables seamless data routing in intermittently-powered battery-free sensing systems. Swift overcomes the challenges posed by device intermittency and heterogeneous energy conditions through three major innovative designs. First, Swift incorporates a reliable node synchronization protocol backed by number theory, ensuring successful synchronization regardless of energy conditions. Second, Swift adopts a low-latency message forwarding protocol, allowing continuous message forwarding without repeated synchronization. Finally, Swift features a simple yet effective mechanism for routing path construction, enabling nodes to obtain the optimal path to the sink node with minimum hops. We implement Swift and perform large-scale experiments representing diverse realworld scenarios. The results demonstrate that Swift achieves an order of magnitude reduction in end-to-end message delivery time compared with the state-of-the-art approaches for intermittentlypowered battery-free sensing systems.

Gaosheng Liu and Lin Wang. IEEE Transactions on Mobile Computing, vol. 23, no. 12, pp. 13406-13419, Dec. 2024, doi: 10.1109/TMC.2024.3429636.

2024

A little certainty is all we need: Discovery and synchronization acceleration in battery-free iot

The vision of sustainable IoT constructed from battery-free devices has attracted ample interest in the research community. Yet, efficient device discovery and synchronization—a fundamental problem in IoT systems—remains a critical challenge mainly due to the uncertain ambient energy availability across battery-free devices. We argue that bringing in a small level of certainty is necessary for facilitating communication in battery-free IoT. We propose Pulsar where we introduce a small number of battery-powered devices, serving as the communication coordinator for a large number of battery-free devices. We develop two communication schemes, namely one-to-one, and all-to-all, for Pulsar. Our results based on simulations and prototype-based experiments show that Pulsar achieves consistently good performance across different scenarios while requiring no special hardware or environmental conditions.ere the description...

Liu, Gaosheng and Nigade, Vinod and Bal, Henri and Wang, Lin. booktitle = {Proceedings of the 8th Asia-Pacific Workshop on Networking}, pages = {38–44}, numpages = {7}, location = {Sydney, Australia}, series = {APNet '24}

2023

Routing for intermittently-powered sensing systems (Best paper award)

Recently, intermittent computing (IC) has received tremendous attention due to its high potential in perpetual sensing for Internet-of-Things (IoT). By harvesting ambient energy, battery-free devices can perform sensing intermittently without maintenance, thus significantly improving IoT sustainability. To build a practical intermittently-powered sensing system, efficient routing across battery-free devices for data delivery is essential. However, the intermittency of these devices brings new challenges, rendering existing routing protocols inapplicable.In this paper, we propose RICS, a new routing scheme tailored for intermittently-powered sensing systems. RICS features two major designs to combat the intermittency challenge, with the goal of achieving low-latency data delivery on a network built with battery-free devices. First, RICS incorporates a fast topology construction protocol for each IC node to establish a path towards the sink node with the least hop count. Second, RICS employs a low-latency message forwarding protocol, which incorporates an efficient synchronization mechanism and a novel technique called pendulum-sync to avoid time-consuming repeated node synchronization. Our evaluation based on an implementation in OMNeT ++ and comprehensive experiments with varying system settings shows that RICS can achieve orders of magnitude latency reduction in data delivery compared with the state-of-the-art.

Gaosheng Liu and Lin Wang. Anaheim, CA, USA, 2023, pp. 274-282, doi: 10.1109/IPCCC59175.2023.10253844.

2021

Self-sustainable cyber-physical systems with collaborative intermittent computing

Write herCyber-physical systems have become a main technology driver for our intelligent society. However, almost all cyber-physical systems rely on battery-powered devices to function, which suffer from high maintenance cost for recharging/replacing the batteries and bring in negative environmental impacts due to the hazardous chemicals used in the batteries. To address this challenge, a new computing paradigm called intermittent computing (IC) was proposed which advocates a battery-free design where cyber-physical devices can be completely powered by energy scavenged from ambient sources such as sunlight, radio waves, and vibrations. Since its advent, many efforts have been made on addressing the challenges in IC, from the hardware to the software stack. In this vision paper, we make a brief summary of existing works on IC and discuss a more realistic setup where, instead of focusing on one IC node as done in most existing works, we propose to build a self-sustainable cyber-physical system through the collaboration of distributed IC devices---collaborative intermittent computing (CIC). We discuss the challenges in CIC and provide a vision for the future cyber-physical systems.e the description...

Gaosheng Liu and Lin Wang. booktitle = {Proceedings of the Twelfth ACM International Conference on Future Energy Systems}, pages = {316–321}, numpages = {6}, series = {e-Energy '21}

2018

[ESRQ: an efficient secure routing method in wireless sensor networks based on q-learning]

Some trust-based methods have been proposed for consideration by guaranteeing the security routing of wireless sensor networks (WSN) recently. However, Most of the current research work usually consumes a lot of energy resources and computing resources or only performs well for specific types of attacks. In this paper, we propose a lightweight and efficient security routing method by combining the trust mechanism with q-learning algorithm. The sensor node determines the credibility of the specific node autonomously and efficiently according to the performance of the behavior of the node. The results show that 1) this method provides a more secure routing mechanism and it can effectively deal with many kinds of attacks; and 2) the method is lightweight and reliable, it provides better energy/packet delivery tradeoff performance compared with other existing trust-based methods and can cope with different conditions.

Gaosheng Liu, Xin Wang, Xiaohong Li, Jianye Hao and Zhiyong Feng. New York, NY, USA, 2018, pp. 149-155, doi: 10.1109/TrustCom/BigDataSE.2018.00032.

WORK EXPERIENCE

15/04/2019 – 01/01/2021 Beijing, China

APPLICATION PROGRAMMER HUAWEI

01/01/2025 – 01/06/2025 Amsterdam, Netherlands

RESEARCHER VU AMSTERDAM

LANGUAGE SKILLS

Mother tongue(s): **CHINESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGL	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HONOURS AND AWARDS

17/11/2023

IPCCC2023 Best Paper Sward – IEEE IPCCC

The best paper award (only one) from IEEE International Conference on Performance, Computing and Communications.

20/10/2024

SenSys2024 Student Travel Grant – ACM SenSys

Student Tavel Grant from ACM Conference on Embedded Networked Sensor Systems. This award encourages young scholars to attend, and communicate with excellent researchers.

2023
Mobisys2023 artifact evaluation committee memeber

2025
Mobisys2025 artifact evaluation committee memeber

SKILLS

C/C++, Python | Office | PCB design | Solder

RECOMMENDATIONS

Henri Bal Professor

- I have known him for 4 years, he is my supervisor

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- I have known him for 4 years, he is my co-supervisor.

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